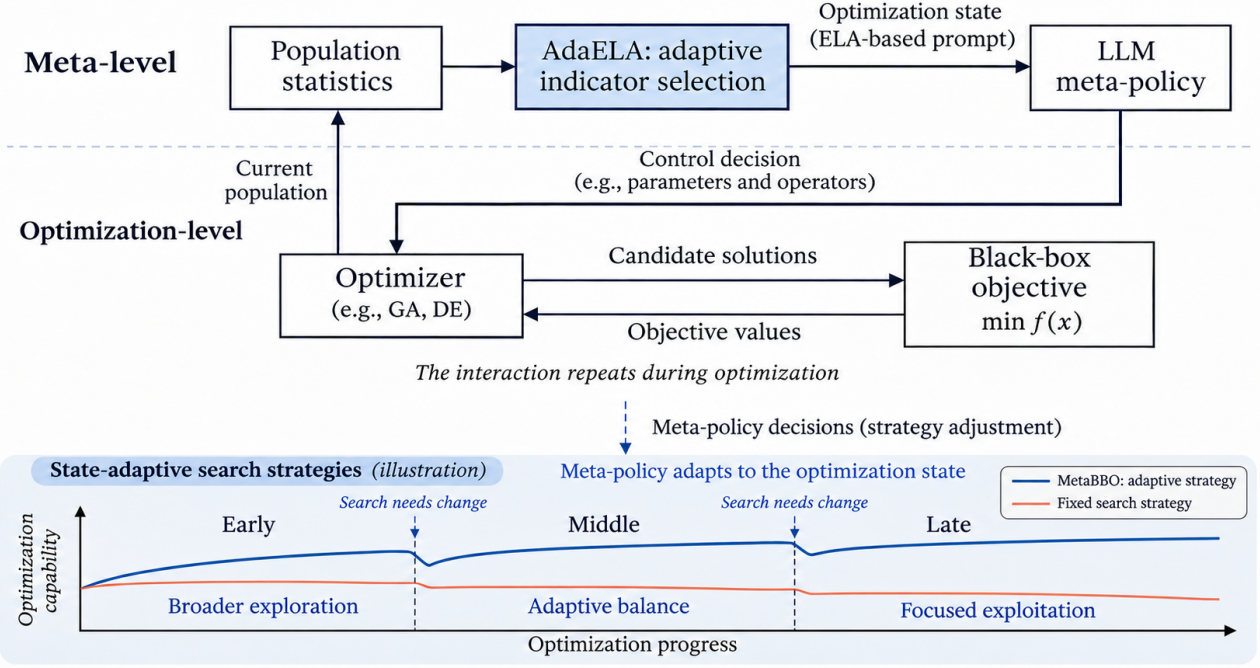

Meta-level
Population statistics
AdaELA: adaptive indicator selection
Optimization state (ELA-based prompt)
LLM meta-policy
Current population
Control decision (e.g., parameters and operators)
Optimization-level
Optimizer (e.g., GA, DE)
Candidate solutions
Objective values
Black-box objective min $f(x)$
The interaction repeats during optimization
Meta-policy decisions (strategy adjustment)
State-adaptive search strategies (illustration)
Meta-policy adapts to the optimization state
MetaBBO: adaptive strategy
Fixed search strategy
Search needs change
Search needs change
Early
Middle
Late
Optimization capability
Broader exploration
Adaptive balance
Focused exploitation
Optimization progress